- **Harmonic numbers**

Proposition C.4. *Let H_n be the n-th harmonic number. Then*

$$H_n = \log(n) + \gamma + \frac{1}{2n} - \sum_{k=1}^{\infty} \frac{B_{2k}}{2k} \frac{1}{n^{2k}} \tag{72}$$

where B_k are the Bernoulli numbers and γ is the Euler-Mascheroni constant.

The Bernoulli numbers are constructed the following way: let $B_k(x)$ be the Bernoulli functions, we define $B_k = B_k(0)$ (and a periodicity constraint). For more informations on Bernoulli numbers, see Casselman introductory article (especially their Fourier series).

Proof. It is a consequence of the Euler-Maclaurin formula (see [Ten22]). Let $k \geq 0$ and $f \in \mathcal{C}^{k+1}([a,b], \mathbf{R})$ with a and b integers, then

$$\sum_{a<m\leq b} f(m) = \int_a^b f(t)\mathrm{d}t + \sum_{0\leq r\leq k} \frac{(-1)^{r+1}B_{r+1}}{(r+1)!}\left(f^{(r)}(b) - f^{(r)}(a)\right) + \frac{(-1)^k}{(k+1)!}\int_a^b B_{k+1}(t) f^{(k+1)}(t)\mathrm{d}t.$$

Choose $a = 1$, $b = n$ and $f(t) = 1/t$. Notice that $f^{(r)}(t) = (-1)^r r!/t^{r+1}$. We then have

$$\begin{aligned}
\sum_{1<m\leq n} \frac{1}{m} &= \log(n) - \sum_{0\leq r\leq k} \frac{B_{r+1}}{r+1}\left(\frac{1}{n^{r+1}} - 1\right) - \int_1^n \frac{B_{k+1}(t)}{t^{k+2}}\mathrm{d}t \\
\sum_{1\leq m\leq n} \frac{1}{m} &= 1 + \log(n) - \frac{B_1}{1}(1/n - 1) - \sum_{1\leq r\leq k} \frac{B_{r+1}}{r+1}\left(\frac{1}{n^{r+1}} - 1\right) - \int_1^n \frac{B_{k+1}(t)}{t^{k+2}}\mathrm{d}t \\
\sum_{1\leq m\leq n} \frac{1}{m} &= \log(n) + \frac{1}{2n} + 1/2 - \sum_{1\leq r\leq k} \frac{B_{r+1}}{r+1}\left(\frac{1}{n^{r+1}} - 1\right) - \int_1^n \frac{B_{k+1}(t)}{t^{k+2}}\mathrm{d}t \\
\sum_{1\leq m\leq n} \frac{1}{m} &= \log(n) + \frac{1}{2n} + 1/2 - \sum_{1\leq r\leq k} \frac{B_{r+1}}{r+1}\frac{1}{n^{r+1}} + \sum_{1\leq r\leq k} \frac{B_{r+1}}{r+1} - \int_1^n \frac{B_{k+1}(t)}{t^{k+2}}\mathrm{d}t \\
\sum_{1\leq m\leq n} \frac{1}{m} &= \log(n) + \frac{1}{2n} + 1/2 - \sum_{1\leq r\leq k} \frac{B_{2r}}{2r}\frac{1}{n^{2r}} + \sum_{1\leq r\leq k} \frac{B_{2r}}{2r} - \int_1^n \frac{B_{k+1}(t)}{t^{k+2}}\mathrm{d}t
\end{aligned}$$

because Bernoulli numbers with odd indices are null. The function f is indefinitely differentiable, thus we can let k tend to infinity and get the desired formula knowing that $\gamma = \frac{1}{2} + \sum_{1\leq r\leq k} \frac{B_{2r}}{2r}$. It ultimately suffices to show that the rest is indeed a rest, *i.e.* it is equal to $\mathcal{O}\left(\frac{1}{n^{k+1}}\right)$. We have the following bound

$$\left|\int_1^n \frac{B_{k+1}(t)}{t^{k+2}}\mathrm{d}t\right| \leq M\left|\int_1^n \frac{1}{t^{k+2}}\mathrm{d}t\right| = M\left|\frac{1}{-(k+1)}\left(\frac{1}{n^{k+1}} - 1\right)\right| = \frac{M}{k+1} - \frac{M}{k+1}\frac{1}{n^{k+1}}$$

because the Bernoulli functions are bounded by their maximum $M := M_k$. □